



SCADA-Based AC Power Forecasting in a Utility-Scale PV Plant: Explainable HistGradientBoosting and Thermal Derating Quantification

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Keywords	Abstract
Photovoltaic (PV) SCADA AC Power Forecasting HistGBDT Thermal Derating	Accurate photovoltaic (PV) power modeling is essential for grid operation, short-term scheduling and plant-level decision making. However, reported performance in the literature varies widely due to differences in feature selection, forecast horizon definition and evaluation protocols. In particular, true exogenous forecasting is often conflated with contemporaneous power tracking, leading to optimistic accuracy estimates with limited operational relevance. This study aims to clarify how feature availability, evaluation design and operational intent shape reported performance in PV power modeling. A leakage-aware and operationally realistic framework is proposed that explicitly distinguishes between exogenous forecasting and operational tracking tasks. Using high-frequency SCADA data from a utility-scale PV plant, two modeling scenarios are evaluated. Scenario A relies exclusively on exogenous meteorological inputs and represents a forecasting-oriented inference task, whereas Scenario B augments these inputs with synchronous electrical current measurements and is interpreted as operational tracking (nowcasting) rather than forward-looking prediction. Multiple regression models, with emphasis on tree-based ensemble methods, are evaluated under transparent and scenario-consistent train-test splitting strategies. Model behavior is examined using post-hoc explainability tools to assess physical consistency and temperature-induced power derating is quantified using a semiparametric regression approach. The results indicate that exogenous-only models recover physically meaningful relationships but exhibit higher numerical errors than current-augmented configurations, reflecting the informational advantage of synchronous measurements rather than improved forecasting capability. Rather than claiming superior numerical accuracy, this work demonstrates how modeling assumptions and evaluation choices directly influence reported performance and its interpretation. By aligning feature availability with operational intent, the proposed framework provides practical guidance for the development, evaluation and deployment of PV power models in real-world applications.

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1. INTRODUCTION

Photovoltaic (PV) power forecasting plays a crucial role in power system operation, short-term scheduling and plant-level decision making. Accurate estimation of PV output supports grid stability and facilitates the integration of renewable energy sources into modern power systems. Despite extensive research efforts, reported forecasting performance in the literature varies significantly, largely due to differences in data sources, feature selection strategies, forecast horizons and evaluation protocols (Antonanzas et al., 2016; Di

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Leo et al., 2025). As a result, direct comparison of model performance across studies remains challenging and methodological clarity has become an increasingly important concern.

In recent years, machine learning (ML) and deep learning (DL) approaches have been widely adopted for short-term PV power forecasting due to their ability to capture nonlinear relationships between meteorological variables and power output. Numerous studies report substantial accuracy improvements over traditional physical or statistical models, particularly for short forecasting horizons (Singla et al., 2021; Di Leo et al., 2025). Among these approaches, advanced architectures such as transformer-based models have attracted growing attention, demonstrating strong predictive capability under carefully designed experimental settings (Piantadosi et al., 2024). Despite these advances, improvements in reported accuracy must be interpreted with caution, particularly when evaluation protocols do not reflect operational constraints.

However, high numerical accuracy alone does not necessarily imply operational reliability. Several studies have highlighted that model performance is often inflated by the inappropriate use of target-adjacent or synchronous input variables, leading to overly optimistic results that lack real-world forecasting relevance (Porawagamage et al., 2024). In particular, the distinction between forward-looking forecasting and contemporaneous power tracking is frequently blurred, despite its critical importance for deployment in operational environments.

This issue is especially relevant in SCADA-based PV modeling, where electrical and meteorological variables are often recorded simultaneously. Models that incorporate synchronous electrical measurements may effectively estimate instantaneous power output, but such configurations correspond to operational tracking (nowcasting) rather than true forecasting if those variables are unavailable at prediction time (Eseye et al., 2018). Failure to explicitly acknowledge this distinction can lead to misinterpretation of model capability and limits the practical value of reported results.

Beyond methodological considerations, physical consistency remains a fundamental requirement for reliable PV power modeling. The negative impact of temperature on PV module performance, commonly referred to as thermal derating, has been well established in the literature. Elevated module temperatures lead to systematic reductions in electrical output, particularly under high irradiance conditions (Skoplaki & Palyvos, 2009). More recent studies further demonstrate that temperature coefficients may vary with irradiance, indicating that temperature effects are more complex than simple linear assumptions suggest (Perin Gasparin et al., 2022).

Against this background, tree-based ensemble methods have emerged as a promising class of models for PV power applications. These methods offer a favorable balance between predictive accuracy, robustness and interpretability and have been successfully applied to large-scale and utility-scale PV forecasting problems (Xie et al., 2025). When combined with explainable artificial intelligence techniques, such as SHAP, tree-based models enable transparent analysis of feature contributions and support the assessment of physical plausibility in data-driven predictions (Fu et al., 2025).

Motivated by these observations, the present study explicitly separates two modeling scenarios based on feature availability and operational intent. The first scenario relies exclusively on exogenous meteorological variables and represents a generalizable inference task suitable for forecasting-oriented applications. The second scenario augments these inputs with electrical current measurements and is interpreted as operational tracking (nowcasting) rather than forward-looking forecasting. This separation enables a transparent assessment of performance gains while avoiding information leakage and misinterpretation.

To address the identified limitations, this study makes the following contributions:

- (i) It introduces a leakage-aware modeling framework that explicitly distinguishes between exogenous inference and operational tracking scenarios.
- (ii) It evaluates multiple regression paradigms under a unified and transparent experimental protocol, enabling fair performance comparison.
- (iii) It integrates post-hoc explainability tools (SHAP and PDP/ICE) to assess the physical consistency of learned relationships.
- (iv) It provides a quantitative estimation of temperature-induced power derating using semiparametric regression, yielding interpretable coefficients with confidence intervals.

2. BACKGROUND AND RELATED WORK

PV power forecasting has been studied extensively because reliable short-term estimates support grid operation, market participation and plant-level decision making. Review papers consistently report that the apparent performance of PV forecasting models varies widely across studies, largely due to differences in forecast horizons, data sources, input feature sets and evaluation protocols (Antonanzas et al., 2016; Sobri et al., 2018; Di Leo et al., 2025). This variability makes cross-study comparison difficult and motivates clearer methodological framing and reporting practices (Inman et al., 2013; Barhmi et al., 2024).

Earlier PV forecasting research often relied on physical or semi-empirical formulations that explicitly connect irradiance and temperature to electrical output. These approaches provide interpretability and physical grounding but typically require detailed system parameters and may underperform when nonlinearities and site-specific effects dominate. In practice, this limitation has accelerated the adoption of data-driven approaches, particularly for short-horizon applications (Antonanzas et al., 2016).

ML models including tree-based ensembles and boosting methods have been widely used due to their ability to capture nonlinear interactions without explicit physical equations (Friedman, 2001; Chen & Guestrin, 2016). Reviews show that these models can be competitive across multiple PV settings, especially when robust preprocessing and careful feature design are applied (Sobri et al., 2018; Di Leo et al., 2025). Deep learning

methods (e.g., transformer-based architectures) have also become prominent, often aiming to improve representation learning in complex spatiotemporal patterns (Piantadosi et al., 2024).

A recurring limitation in the literature, however, concerns evaluation realism and feature availability at prediction time. Many studies report high accuracy while using electrical variables measured synchronously with the target (e.g., currents or power-adjacent signals). Although this can improve numerical metrics, it changes the task definition from forward-looking forecasting to contemporaneous tracking/nowcasting if those inputs are not available (or not intended to be used) ahead of time. This concern is closely related to methodological best practices in solar forecasting, where transparent protocols and time-consistent validation are emphasized to avoid misleading conclusions (Inman et al., 2013; Barhmi et al., 2024). More broadly, for time-indexed prediction tasks, inappropriate validation strategies can bias performance estimates; time-series-aware validation and careful split design are therefore recommended (Bergmeir et al., 2018).

In parallel with accuracy-focused research, there is increasing interest in interpretability and physical consistency. Post-hoc explainability techniques, such as SHAP, enable researchers to audit whether learned relationships match known PV behavior and whether models rely on physically plausible drivers (Lundberg & Lee, 2017; Fu et al., 2025). Nevertheless, interpretability is often presented as a secondary add-on rather than integrated into the evaluation logic (Di Leo et al., 2025).

Another important theme is temperature-related performance loss (thermal derating). The negative impact of module temperature on PV electrical performance is well established and review work provides a clear physical basis for expected trends (Skoplaki & Palyvos, 2009). More recent empirical studies further show that temperature coefficients can vary with irradiance, indicating that temperature effects may be more nuanced than simple linear assumptions (Perin Gasparin et al., 2022). Despite this, many data-driven PV forecasting studies still treat temperature effects only qualitatively or through broad correlations, limiting practical interpretability for operational diagnostics (Di Leo et al., 2025).

Overall, the literature demonstrates strong progress in model development but highlights persistent gaps in (i) clearly separating forecasting from tracking tasks when synchronous electrical variables are used, (ii) applying evaluation protocols that respect time dependence and operational constraints and (iii) integrating interpretability to assess physical consistency alongside predictive performance (Inman et al., 2013; Barhmi et al., 2024; Di Leo et al., 2025). The present study addresses these gaps by adopting a leakage-aware, operationally realistic evaluation framework and by pairing predictive assessment with explainability and temperature-derating analysis.

3. DATA AND PRE-PROCESSING

3.1. Dataset and Variables

The dataset used in this study consists of supervisory control and data acquisition (SCADA) measurements collected from an operational utility-scale PV power plant located near Hassan, Karnataka, India. Data were recorded using an on-site data logger installed at the plant and represent normal operating conditions of a grid-connected PV facility (Eseye et al., 2018; Bööke & Lindfors, 2020).

The PV plant has a nominal installed capacity of approximately 350 kWp. The maximum observed AC power output in the dataset is approximately 333 kW, which is consistent with inverter-rated capacity, DC-to-AC conversion limits and standard operational clipping behavior. This confirms the physical plausibility of the recorded power values.

The dataset comprises 118,865 synchronized records and includes both meteorological and electrical variables. Meteorological inputs consist of plane-of-array irradiance (IRR, $\text{W}\cdot\text{m}^{-2}$), module back-surface temperature ($^{\circ}\text{C}$), ambient temperature ($^{\circ}\text{C}$) and wind speed ($\text{m}\cdot\text{s}^{-1}$). Electrical measurements include DC current (A) and three-phase AC currents (I_r , I_y , I_b) measured at the inverter output, along with the corresponding AC power output (kW), which serves as the prediction target.

All variables were recorded at uniform sampling intervals and time-aligned by the data logger. The dataset was inspected for missing values, duplicate entries and nonphysical measurements. No missing or duplicated records were detected after basic screening.

All numeric values are reported using international decimal-point notation. Apparent irradiance values exceeding $1000 \text{ W}\cdot\text{m}^{-2}$ correspond to clear-sky conditions and tilted module geometry and are physically consistent with utility-scale PV operation. Module temperature values up to approximately 72°C are also within expected operating limits for crystalline-silicon PV modules under high irradiance conditions (Skoplaki & Palyvos, 2009).

The dataset is publicly available via Kaggle and was independently examined for internal consistency and unit correctness prior to analysis. While the authors did not directly collect the raw measurements, the data originate from a real operational PV installation rather than a simulated environment, making them suitable for evaluating practical SCADA-based modeling frameworks (Kanagolkar, n.d.).

Two distinct feature configurations are derived from the dataset in accordance with the study objectives. The first configuration includes only exogenous meteorological variables and is designed to represent a generalizable inference setting without target-adjacent information. The second configuration augments these inputs with electrical current measurements and reflects an operational tracking (nowcasting) scenario rather than forward-looking forecasting. This separation enables a transparent assessment of performance gains while avoiding misinterpretation due to information leakage (Alcañiz et al., 2023).

The dataset, publicly available on Kaggle (Kanagolkar, n.d.), is summarized in Table 1.

Table 1. Descriptive statistics of nine variables ($N = 118,865$)

	Unit	mean	std	min	25%	50%	75%	max	kW
MODULE_TEMP	°C	37.14	12.02	8.86	26.70	36.46	46.68	72.45	37.14
Amb_Temp	°C	22.96	3.92	10.41	20.08	23.06	26.03	34.96	22.96
WIND_Speed	$\text{m}\cdot\text{s}^{-1}$	2.24	2.30	0.24	0.28	0.59	4.43	5.97	2.24
IRR (Irradiance (POA))	$\text{W}\cdot\text{m}^{-2}$	428.09	313.28	2.13	155.45	357.74	687.80	1494.85	428.09
DC Current	A	355.90	264.86	0.60	126.18	298.60	576.32	995.68	355.90
AC Ir	A	172.33	120.51	1.40	65.60	150.00	277.70	461.20	172.33
AC Iy	A	172.19	120.45	1.40	65.50	150.00	277.40	461.10	172.19
AC Ib	A	172.35	120.50	1.50	65.60	150.10	277.60	461.70	172.35
AC Power	kW	128.08	91.19	0.39	47.74	110.67	206.59	332.61	128.08

Table 1 reports summary statistics (mean, standard deviation, minimum, quartiles and maximum) for all continuous variables. All values are computed directly from the original dataset and reported using international decimal-point notation. AC power is expressed in kW. The dataset contains no missing or duplicate records.

3.2. Cleaning and Basic Screening

Prior to model development, the dataset was subjected to a structured cleaning and basic screening procedure aimed at ensuring numerical consistency, physical plausibility and alignment with operational conditions. As established in Section 3.1, the dataset contains no missing or duplicate records; therefore, no imputation or record removal was required. The focus of this stage was not aggressive data filtering, but rather diagnostic examination to support transparent and reproducible modeling decisions.

All variables in the dataset are numeric and unit-consistent. The only unit conversion applied was the transformation of AC power from watts to kilowatts to standardize reporting across experiments. No scaling, normalization, or target-dependent transformations were applied during this stage, in order to preserve the original physical meaning of the measurements and avoid unintended information leakage.

Univariate distributions of all continuous variables were examined using histograms and summary statistics. Irradiance and power-related variables exhibit right-skewed distributions consistent with the diurnal nature of photovoltaic generation, while temperature-related variables show unimodal distributions within physically plausible operating ranges. These distributional characteristics informed the selection of tree-based regression models, which are robust to skewed inputs and do not rely on strict distributional assumptions.

Bivariate relationships between key input variables and AC power output were explored through scatter plots. Plane-of-array irradiance demonstrates a strong monotonic relationship with AC power, confirming its role as the dominant exogenous driver. Module temperature shows a negative association with power output at higher irradiance levels, reflecting well-established thermal derating effects. Wind speed and ambient temperature exhibit weaker but physically interpretable relationships, primarily through their indirect influence on module temperature.

To assess multicollinearity and feature redundancy, a correlation heatmap was constructed. Electrical current measurements show strong correlations with AC power due to their synchronous measurement, whereas meteorological variables display moderate intercorrelations. These observations directly motivated the explicit separation between exogenous-only feature configurations and current-augmented configurations, as discussed in Section 3.1 and form the basis of the leakage-aware experimental design adopted in this study.

For exploratory visualization purposes only, a daytime subset defined by plane-of-array irradiance exceeding $50 \text{ W}\cdot\text{m}^{-2}$ ($\text{IRR} > 50$) was used where explicitly stated. This threshold was employed solely to exclude nighttime conditions in diagnostic plots and was not used in model training, evaluation, or performance reporting. Screening choices were guided by operational realism and transparency rather than performance optimization.

Overall, the cleaning and basic screening stage functioned as a diagnostic step linking data characteristics to methodological choices, rather than as a feature elimination or optimization procedure. The insights obtained from distributional analysis, pairwise relationships and correlation structure directly informed the selection of model classes, feature configurations and interpretability strategies employed in subsequent sections.

3.3. Feature Engineering and Exploratory Structure

To summarize three-phase loading with a single interpretable descriptor, we define the phase-average AC current;

$$I_{avg} = (AC I_r + AC I_y + AC I_b)/3 \quad (1)$$

This variable is used consistently throughout the analysis for diagnostics and sanity checks (Alcañiz et al., 2023). To preserve the interpretability of contemporaneous driver effects, calendar and lagged features are deliberately excluded from the baseline configuration (Singla et al., 2021). Immediately following the descriptive statistics, univariate distributions of all variables are examined to assess range, skewness and potential outliers (Figure 1).

Violin contours depict the overall distribution shape, while inner boxes represent medians and interquartile ranges. Irradiance and phase currents exhibit pronounced right-skewed tails, whereas temperature variables

show narrower distributions. AC Power is moderately right-skewed, consistent with daytime PV operation (Figure 2).

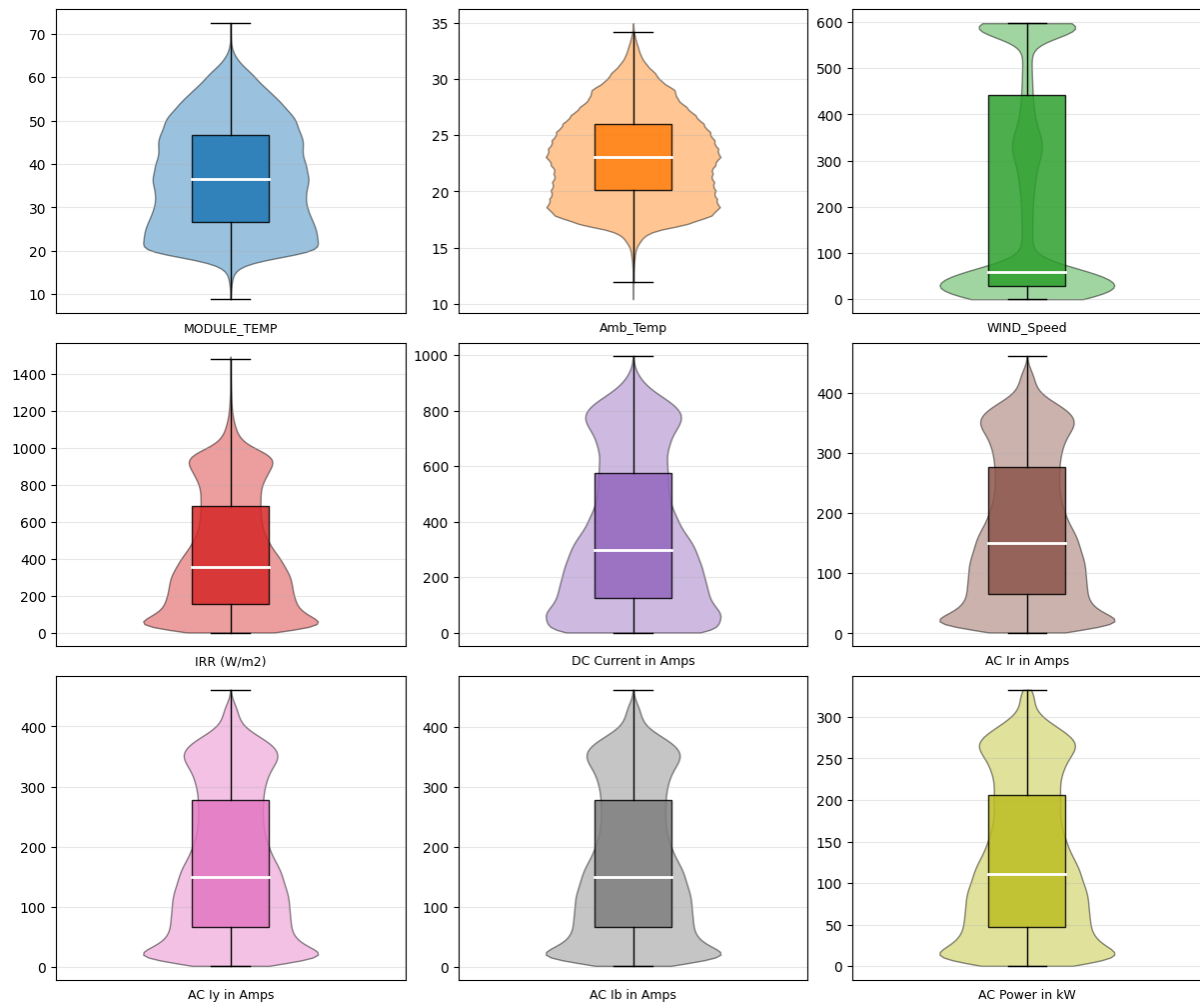


Figure 1. Univariate distributions for nine variables

Next, bivariate relationships are explored to identify dominant effects and potential nonlinear interactions.

Panels include:

AC Power vs IRR, AC Power vs MODULE_TEMP, AC Power vs Amb_Temp, AC Power vs DC Current, AC Power vs Iavg, AC Power vs WIND_Speed, WIND_Speed vs Amb_Temp, IRR vs MODULE_TEMP and DC Current vs IRR.

Irradiance and current variables show strong positive coupling with AC Power, while wind speed exhibits only a weak correlation (Alcañiz et al., 2023).

Finally, linear associations among all continuous variables are summarized using a Pearson correlation heatmap (Figure 3).

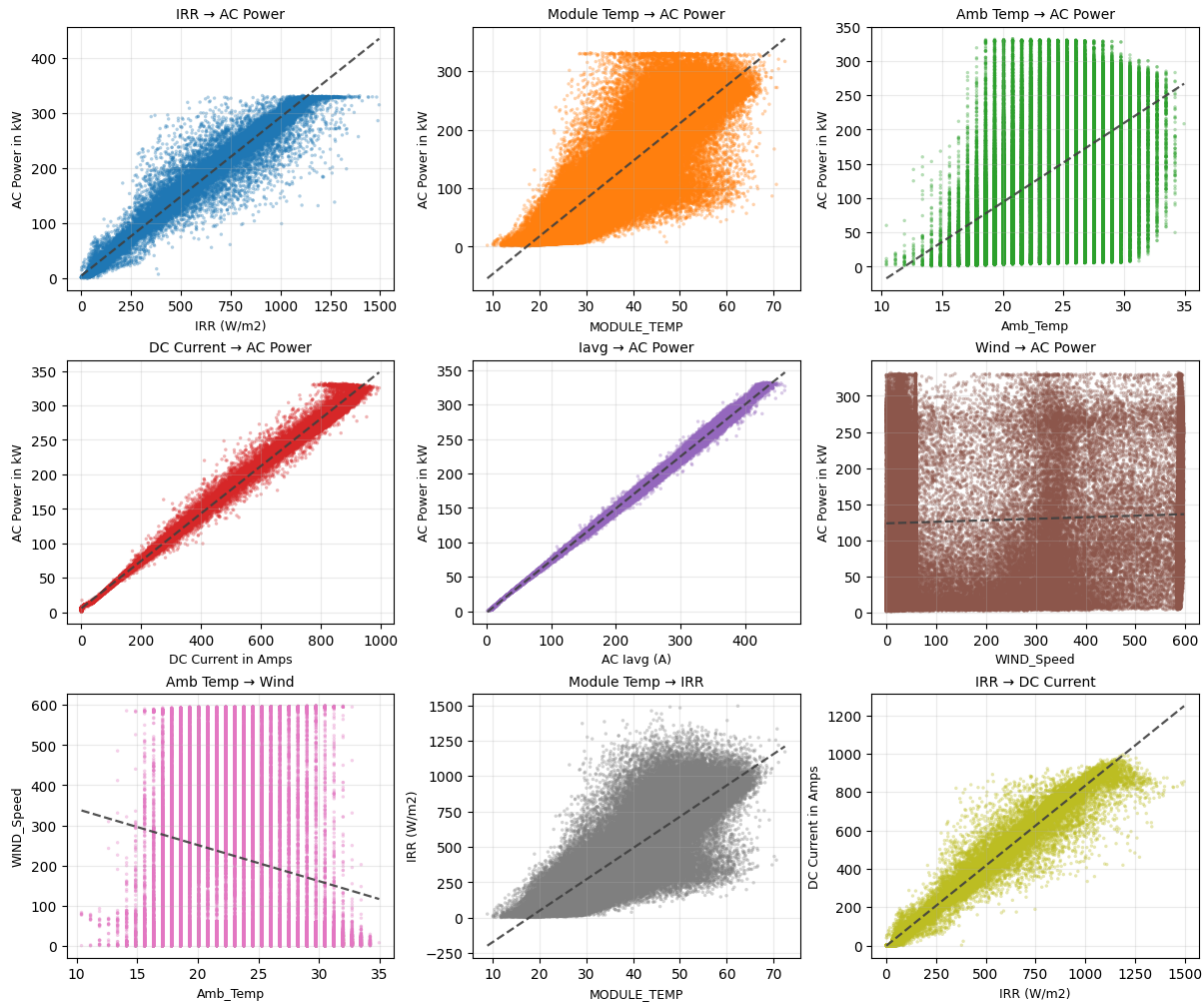


Figure 2. Scatterplots with neutral dashed OLS trend lines

AC Power correlates strongly with irradiance and electrical currents ($|r| \approx 0.99\text{--}1.00$). MODULE_TEMP co-varies with irradiance and power ($r \approx 0.85$), while WIND_Speed shows weak association ($|r| \leq 0.15$) (Singla et al., 2021). These findings confirm irradiance as the dominant exogenous driver and provide a clear basis for the two modeling scenarios introduced in Section 4.

4. METHODS

4.1. Problem Formulation and Modeling Scenarios

Let y_t denote the measured AC power (in kW) output of the PV plant at time index t and let X_t represent a vector of input features available at the same time index. The objective of this study is to model the functional relationship between input measurements and AC power output under different assumptions regarding feature availability and operational intent.

A key methodological distinction is made between forward-looking forecasting and contemporaneous power estimation (Figure 4). In practical PV operations, some variables are available only ex post or are measured synchronously with the target output, which fundamentally alters the interpretation of model predictions. To

avoid ambiguity and information leakage, two modeling scenarios are explicitly defined and evaluated separately (Inman et al., 2013; Barhmi et al., 2024; Porawagamage et al., 2024).



Figure 3. Pearson correlation heatmap across nine variables

4.1.1. Scenario A: Exogenous Inference (Forecast-Oriented Setting)

In Scenario A, the model is formulated as

$$\hat{y}_t = f(X_t^{\text{exo}}) \quad (2)$$

where X_t^{exo} consists exclusively of exogenous meteorological variables, including plane-of-array irradiance, module temperature, ambient temperature and wind speed. No electrical measurements that are directly coupled to the power output are included in this configuration.

This scenario represents a generalizable inference task aligned with forecasting-oriented applications, in which only environmental inputs are assumed to be available. By construction, Scenario A avoids target-adjacent information and serves as a leakage-safe baseline for evaluating model performance under realistic forecasting constraints (Bergmeir et al., 2018; Porawagamage et al., 2024).

4.1.2. Scenario B: Operational Tracking (Nowcasting Setting)

In Scenario B, the input feature set is augmented with electrical current measurements recorded synchronously with the target output:

$$\hat{y}_t = f(X_t^{\text{exo}}, X_t^{\text{elec}}) \quad (3)$$

where X_t^{elec} includes DC current and three-phase AC currents measured at the inverter level.

This configuration does not represent forward-looking forecasting, but rather operational tracking (nowcasting), as the additional electrical inputs contain contemporaneous information that is intrinsically linked to the instantaneous power output (Porawagamage et al., 2024). Scenario B is therefore evaluated separately and interpreted as an upper bound on achievable performance under real-time SCADA monitoring conditions.

4.1.3. Interpretation and Scope;

The explicit separation between Scenario A and Scenario B is central to the experimental design of this study. Performance differences between the two scenarios are not interpreted as improvements in forecasting capability, but as quantification of the informational advantage provided by synchronous electrical measurements. This distinction ensures that reported results remain physically meaningful, operationally realistic and free from unintended information leakage (Inman et al., 2013; Barhmi et al., 2024; Porawagamage et al., 2024).

All models evaluated in subsequent sections are trained and tested independently under each scenario using identical evaluation protocols. This design enables a transparent comparison of modeling paradigms while preserving the interpretability and validity of performance metrics across different operational contexts (Bergmeir et al., 2018).

The diagram illustrates the complete pipeline for SCADA-based PV AC power forecasting ($N = 118,865$, nine numeric variables), encompassing five major stages:

- Data ingestion and preprocessing – duplicate removal, unit alignment and daytime filtering ($\text{IRR} > 50 \text{ W} \cdot \text{m}^{-2}$).
- Feature engineering – calculation of mean three-phase current (I_{avg}).
- Scenario-based modeling – comparison of two feature regimes (A: exogenous-only, B: exogenous + electrical currents).
- Model evaluation – six regressors (HistGBDT, XGBoost, Linear, SVR, Decision Tree, Tweedie GLM) assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2), Mean Absolute Percentage Error (MAPE) and Symmetric Mean Absolute Percentage Error (sMAPE), excluding low-power cases ($\leq 0.5 \text{ kW}$).

- Post-hoc analysis – model explainability via SHAP, PDP/ICE and thermal derating quantification through high-irradiance OLS and semiparametric estimation of β_T .

Among the evaluated models, HistGBDT is highlighted as a strong candidate due to its balance between accuracy, efficiency and interpretability and is therefore selected as the primary reference model in subsequent analyses.

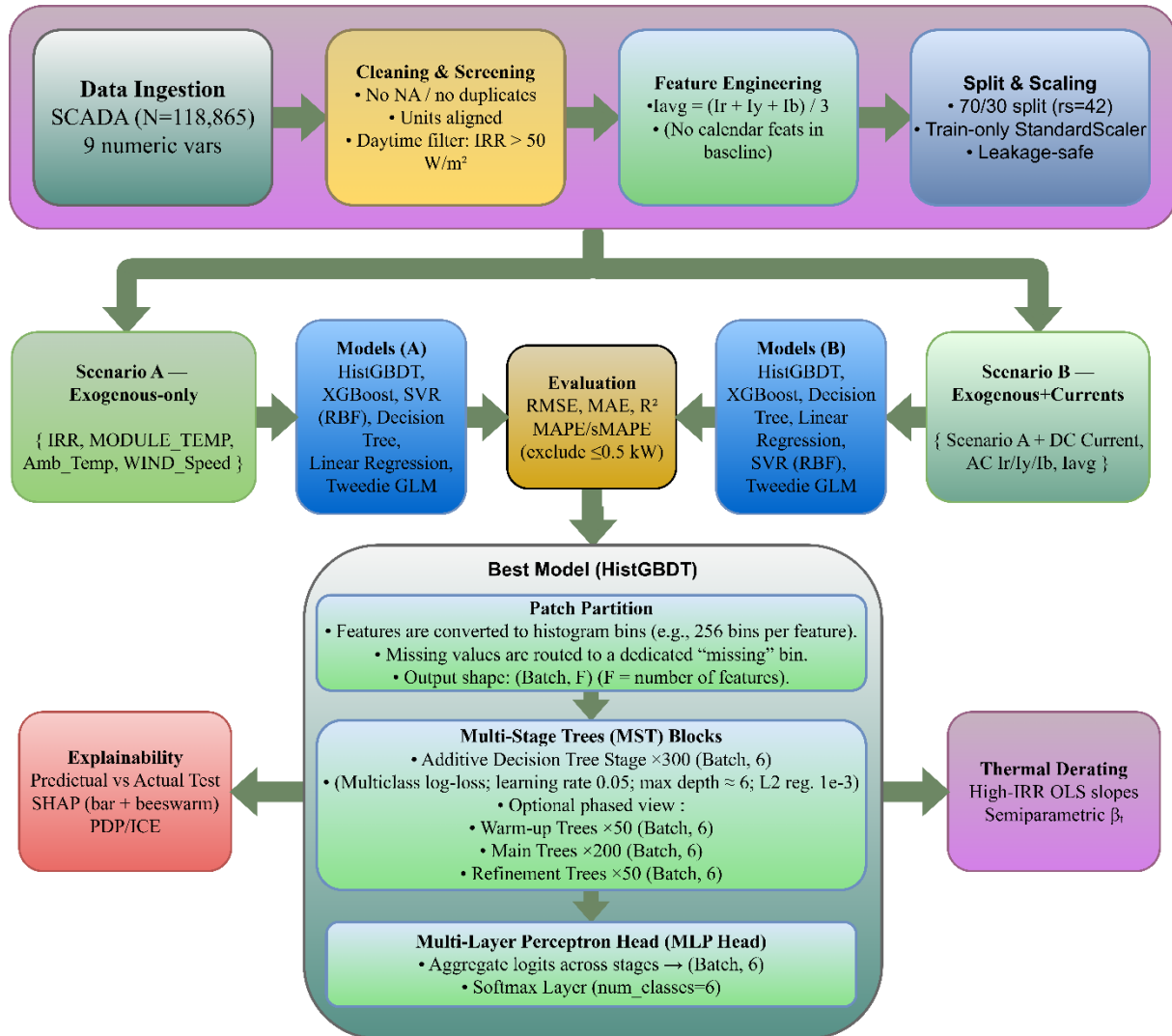


Figure 4. Overall methodological workflow of the proposed PV forecasting framework

4.2. Train–Test Split and Preprocessing

The dataset used in this study consists of time-indexed SCADA measurements collected under continuous PV plant operation. Given the strong temporal structure and diurnal patterns inherent in PV power generation, the choice of train–test splitting strategy plays a critical role in ensuring valid and interpretable performance evaluation. The overall data partitioning strategy, preprocessing choices and evaluation protocol adopted in this study are summarized in Table 2.

Table 2. Train–test split and evaluation protocol summary

Item	Setting
Dataset size	N = 118,865 observations; nine numeric features + AC power target
Data type	Time-indexed SCADA measurements from an operational PV plant
Daytime filter (EDA & modeling)	Plane-of-array irradiance $IRR > 50 \text{ W} \cdot \text{m}^{-2}$ (used for visualization only)
Scenario A split (forecast-oriented)	Temporal split: first 70% of observations for training, remaining 30% for testing (chronological order preserved)
Scenario B split (operational tracking)	Random split: 70% training, 30% testing; shuffle = True; random_state = 42
Feature scaling	No scaling or normalization applied to preserve physical units
Leakage control	Scenario-specific feature separation; no target-adjacent variables used in Scenario A; all preprocessing decisions independent of test data
Scenarios	A: Exogenous-only {IRR, MODULE_TEMP, Amb_Temp, WIND_Speed} (forecasting) B: Exogenous + electrical currents {DC, AC Ir/Iy/Ib} (nowcasting)
Percentage-error handling	Samples with AC power $\leq 0.5 \text{ kW}$ excluded from MAPE and sMAPE computation to avoid numerical instability
Reported metrics	RMSE, MAE, R^2 (full test set); MAPE, sMAPE (filtered subset)

To maintain consistency with the modeling objectives defined in Section 4.1, different data-splitting strategies were employed for each scenario. For Scenario A (exogenous inference), a temporal split was applied, whereby the first 70% of observations in chronological order were used for model training and the remaining 30% for testing. This approach reflects realistic forecasting conditions, in which future observations are unavailable at training time and prevents information leakage arising from random shuffling in time-series data (Bergmeir et al., 2018).

In contrast, Scenario B (operational tracking) aims to estimate contemporaneous power output using synchronous electrical measurements rather than to perform forward-looking prediction. In this setting, temporal causality is not violated by random partitioning, as inputs and targets are observed simultaneously. Accordingly, a random 70:30 train–test split with shuffling enabled (random_state = 42) was adopted for Scenario B to assess generalization performance under stationary operational conditions (Porawagamage et al., 2024).

The 70:30 split ratio was selected as a balance between training data sufficiency and evaluation robustness, given the large sample size of the dataset. Sensitivity checks using alternative split ratios (e.g., 80:20) yielded comparable performance trends and are therefore not reported separately for brevity.

Model performance was evaluated using a consistent set of error metrics across all experiments, including RMSE, MAE and R^2 all computed on the full test sets. Percentage-based metrics, including MAPE and symmetric MAPE (sMAPE), were calculated on a filtered subset excluding samples with AC power less than or equal to 0.5 kW to avoid numerical instability under near-zero denominators (Hyndman & Koehler, 2006). Details of metric definitions and handling of percentage errors are provided in Table 3.

No scaling or normalization was applied to the input variables during preprocessing in order to preserve their physical units and maintain interpretability. All preprocessing decisions were made independently of the test data and were aligned with the scenario-specific feature configurations described in Section 4.1.

Table 2 is positioned immediately after this subsection to provide a concise summary of the train–test splitting strategies, preprocessing assumptions, scenario definitions and evaluation metrics used throughout the study.

Model performance is assessed using RMSE, MAE, R^2 , MAPE and sMAPE. RMSE, MAE and R^2 are computed over the full test set, while percentage-based metrics are reported only for samples with AC power greater than 0.5 kW to ensure numerical stability and meaningful interpretation. This exclusion criterion prevents division-by-zero and inflated percentage errors at low power levels (Hyndman & Koehler, 2006). Definitions and references for all metrics are provided in Table 3.

Table 3. Metrics and percentage-error handling

Metric	Definition	Domain/Handling	Reference
RMSE	$\sqrt{\frac{1}{m} \sum_{j=1}^m (y_j - \hat{y}_j)^2}$	Computed on full test set	(Willmott & Matsuura, 2005)
MAE	$\frac{1}{m} \sum_{j=1}^m y_j - \hat{y}_j $	Computed on full test set	(Willmott & Matsuura, 2005)
R^2	$1 - \frac{\sum_{j=1}^m (y_j - \hat{y}_j)^2}{\sum_{j=1}^m (y_j - \bar{y})^2}$	Computed on full test set	(Draper & Smith, 2014)
MAPE(%)	$\frac{100}{I} \sum_{j \in I} \left \frac{y_j - \hat{y}_j}{y_j} \right $	Percentage errors computed on subset $I = \{i: y_i > 0.5 \text{ kW}\}$	(Hyndman & Koehler, 2006)
sMAPE(%)	$\frac{100}{I} \sum_{j \in I} \frac{ y_j - \hat{y}_j }{(y_j + \hat{y}_j)/2}$	Same subset rule as MAPE; values in %	(Hyndman & Koehler, 2006)

4.3. Models and Hyperparameter Optimization

To systematically assess the impact of modeling assumptions and feature availability on PV power estimation, multiple regression algorithms were evaluated in this study. The selected models span different levels of complexity and inductive bias, enabling a balanced comparison between predictive performance, robustness and interpretability under the two modeling scenarios defined in Section 4.1.

The evaluated model set includes linear regression (OLS), support vector regression with a radial basis function kernel (SVR–RBF), decision tree regression (CART), Tweedie generalized linear models (GLM), gradient-boosting-based ensemble methods and a shallow neural network baseline (Smola & Schölkopf, 2004; Dunn & Smyth, 2005; Breiman et al., 2017). These models were selected to represent widely adopted regression

paradigms in PV power modeling rather than to introduce architectural novelty. Linear and GLM-based models provide transparent baselines, while nonlinear tree-based ensembles and neural networks capture complex interactions between meteorological and electrical variables without extensive feature engineering.

Particular emphasis is placed on gradient-boosting-based models due to their favorable trade-off between predictive accuracy and interpretability. Among these, the histogram-based gradient boosting decision tree (HistGBDT) model is treated as the primary model in subsequent analyses owing to its computational efficiency, tolerance to skewed feature distributions and compatibility with post-hoc explainability techniques (Friedman, 2001). Extreme Gradient Boosting (XGBoost) is included as a complementary boosting framework to benchmark performance consistency across implementations (Chen & Guestrin, 2016).

To address Reviewer 3's request for a deep learning baseline, a Multi-Layer Perceptron (MLP) regressor was additionally evaluated. The MLP was intentionally kept shallow and was applied to the same tabular SCADA inputs used by the other models. This design choice reflects the snapshot-based nature of the dataset and allows the performance of a neural network to be assessed under fair and comparable conditions, without introducing sequence-specific architectures that are not central to the study's objectives.

Hyperparameter optimization was conducted separately for each model using only the training data to avoid information leakage. For most models, randomized search was employed to efficiently explore the hyperparameter space under practical computational constraints, while grid search was applied where appropriate (e.g., Tweedie GLM). Model tuning was performed using five-fold cross-validation applied exclusively to the training sets. For Scenario A, cross-validation respected the temporal ordering of the data to remain consistent with the forecasting-oriented setting, whereas standard k-fold cross-validation was applied for Scenario B, reflecting the stationary nature of the operational tracking task.

The MLP model was trained within a standardized preprocessing pipeline that includes feature scaling, as neural networks are sensitive to input magnitude. Early stopping was employed using an internal validation split within the training set to mitigate overfitting. All preprocessing steps, hyperparameter tuning and early stopping procedures were confined strictly to the training data.

The hyperparameter tuning strategy, validation approach and final selected hyperparameter values for all evaluated models are summarized in Table 4, which is positioned immediately after this subsection. All reported hyperparameters correspond to the best-performing configurations identified during cross-validation and were fixed prior to final model evaluation on the held-out test sets defined in Section 4.2.

All models were implemented using standard open-source machine learning libraries to ensure reproducibility. Unless otherwise stated, default loss functions were used. Final model performance was assessed exclusively on the test sets and no test data were used at any stage of model selection or tuning.

Table 4. *Optimized hyperparameters and tuning strategy for all models*

Model	Tuning method	Validation strategy	Key hyperparameters
HistGradientBoosting (primary)	Randomized search	5-fold CV on training set only	learning_rate = 0.06; max_iter = 600; max_depth = None; l2_regularization = 1.0; random_state = 42
XGBoost	Randomized search	5-fold CV on training set only	n_estimators = 600; learning_rate = 0.05; subsample = 0.8; colsample_bytree = 0.8; random_state = 42; n_jobs = -1
Linear Regression (OLS)	No tuning (closed-form)	Not applicable	fit_intercept = True
SVR (RBF)	Randomized search	5-fold CV on training set only	kernel = RBF; C = 1.0; gamma = "scale"; epsilon = 0.1
Decision Tree (CART)	Randomized search	5-fold CV on training set only	max_depth = 10; min_samples_split = 5; random_state = 42
Tweedie GLM	Grid search	5-fold CV on training set only	power = 1.5; link = log; alpha = 5e-4; max_iter = 10,000
Multi-Layer Perceptron (MLP)	Manual tuning + early stopping	Validation split within training set	hidden layers = (64, 32); activation = ReLU; optimizer = Adam; learning_rate_init = 1e-3; max_epochs = 500; early_stopping = True

4.4. Metrics and Evaluation Protocol

Model performance was evaluated using a set of widely adopted regression metrics, including RMSE, MAE and the R^2 , which are standard in energy forecasting studies (Willmott & Matsuura, 2005). Percentage-based metrics (MAPE and sMAPE) were computed only for samples with AC power exceeding 0.5 kW to avoid numerical instability and inflated errors under near-zero denominators, following established recommendations in the forecasting literature (Hyndman & Koehler, 2006). Definitions and references for all metrics are summarized in Table 4.

Metric definitions follow standard formulations in regression and forecasting literature. Percentage-based metrics exclude samples with AC power ≤ 0.5 kW to ensure numerical stability (Hyndman & Koehler, 2006).

4.5. Explainability and Model Interpretation

To ensure interpretability and transparency of the data-driven models, **SHAP values** are computed to quantify both global and local feature attributions (Kuzlu et al., 2020). SHAP-based global importance rankings and beeswarm visualizations are reported in the Results section to identify dominant drivers of AC power output and to assess the consistency of learned relationships across the feature space.

In addition to SHAP, Partial Dependence Plots (PDP) and Individual Conditional Expectation (ICE) curves are employed to examine marginal effects and potential nonlinear interactions between input variables and predicted power output. PDPs provide an aggregated view of the average model response to individual features, while ICE curves reveal instance-level variability around these averages, enabling detection of heterogeneous effects that may be masked in global summaries (Friedman, 2001).

Together, these explainability tools support a physically grounded interpretation of model behavior by verifying whether learned dependencies align with established photovoltaic performance principles, such as

irradiance dominance and temperature-induced efficiency losses. Therefore, the combined use of SHAP, PDP and ICE facilitates transparent model auditing and enhances the practical interpretability of the proposed modelling framework.

4.6. Thermal Derating Analysis

Temperature-induced derating represents a critical nonlinearity in PV system behavior, directly affecting energy yield and operational reliability. To quantify thermal sensitivity in a manner that is both statistically robust and physically interpretable, two complementary approaches are employed to link data-driven predictions with established PV performance physics.

- High-irradiance OLS fitting: Ordinary Least Squares (OLS) regressions are applied within discrete irradiance bands (600–1000 W·m⁻²) to estimate band-specific temperature coefficients, β_T (kW·°C⁻¹) together with 95% confidence intervals.

Restricting the analysis to high-irradiance conditions reduces confounding effects associated with low-power operation and enables localized assessment of temperature sensitivity under near-nominal operating regimes (Draper & Smith, 2014).

- Semiparametric modeling: To obtain a unified estimate of temperature-induced derating while accounting for nonlinear irradiance effects, a semiparametric regression model is employed, expressed as:

$$AC\ Power = f(IRR) + \beta_T \cdot MODULE_TEMP + \varepsilon \quad (4)$$

where $f(\cdot)$ denotes a smooth spline function of irradiance, β_T is the partial temperature coefficient and ε represents the residual error term. This formulation isolates the net temperature effect after controlling for irradiance and yields a globally interpretable estimate of thermal derating, summarized alongside the model coefficient of determination (R^2) (Nan et al., 2024).

By combining localized OLS-based estimates with a global semiparametric formulation, this dual approach enables cross-validation of temperature effects across operating regimes and provides a physically consistent quantification of thermal derating suitable for diagnostic analysis and operational decision support.

5. RESULTS AND DISCUSSION

5.1. Exploratory Analysis

The distributions and interrelationships of meteorological and electrical variables were examined to identify dominant factors influencing AC power generation. As established in the exploratory analysis presented in Section 3.3, irradiance and three-phase electrical currents exhibit pronounced right-skewed distributions, whereas module and ambient temperatures span narrower operating ranges. The target variable, AC power

output, is similarly right-skewed, reflecting typical utility-scale PV operation under variable irradiance conditions.

Further analysis confirms that AC power is strongly associated with irradiance and electrical current measurements, while wind speed shows only weak correlation. These relationships are consistent with established PV physics, where irradiance is the primary driver of power variability and temperature acts as a secondary modulating factor. Correlation analysis indicates near-perfect association between AC power and irradiance and phase currents ($|r| \approx 0.99\text{--}1.00$), a strong positive co-variation between module temperature and irradiance ($r \approx 0.85$) and negligible association with wind speed ($|r| \leq 0.15$).

These findings have direct implications for model design and evaluation. The strong dependence of AC power on irradiance alone motivates the formulation of Scenario A, which relies exclusively on exogenous meteorological inputs and represents a leakage-safe forecasting setting. Conversely, the near-deterministic relationship between electrical currents and power output justifies Scenario B, where current measurements are included and interpreted as operational tracking rather than forward-looking prediction.

This distinction further motivates the joint reporting of absolute error metrics (RMSE, MAE) and percentage-based metrics (MAPE, sMAPE) under a leakage-aware protocol, ensuring that model performance is evaluated consistently with feature availability and operational intent.

5.2. Scenario A: Exogenous-only Performance

Using only meteorological inputs (plane-of-array irradiance, module temperature, ambient temperature and wind speed), tree-based ensemble models provide the most favorable balance between predictive accuracy, generalization and interpretability (Wang et al., 2018; Xie et al., 2025).

As summarized in Table 5, the HistGBDT model achieves the lowest overall error (RMSE = 11.75 kW, MAE = 6.52 kW) with an R^2 of 0.983, establishing a strong baseline for SCADA-based PV forecasting without electrical telemetry. XGBoost performs comparably (RMSE = 11.83 kW, $R^2 = 0.983$) and yields the lowest percentage errors (MAPE = 8.96%). Linear regression, SVR and decision tree models exhibit slightly higher errors (RMSE $\approx 12\text{--}14$ kW), while the Tweedie GLM baseline deviates substantially (RMSE ≈ 42.4 kW, $R^2 = 0.785$), reflecting its limited flexibility under nonlinear irradiance–temperature interactions (Di Leo et al., 2025).

The MLP baseline achieves moderate predictive performance (RMSE = 12.64 kW, $R^2 = 0.981$). While this confirms that shallow neural networks can capture nonlinear patterns in the data, their performance remains inferior to that of gradient-boosting ensembles for tabular, snapshot-based SCADA inputs (Di Leo et al., 2025).

As illustrated in Figure 5, predicted versus actual AC power values for the HistGBDT model align closely along the 45° reference line, confirming high goodness of fit and stable generalization under a temporally consistent forecasting setup.

Table 5. Scenario A (Exogenous-only) test performance

Scenario	Model	RMSE	MAE	MAPE (%)	sMAPE (%)	R ²
Exogenous-only	HistGBDT	11.750	6.516	9.1519	8.2076	0.983441
Exogenous-only	XGBoost	11.827	6.521	8.961	8.138	0.983225
Exogenous-only	SVR (RBF)	12.108	6.667	10.076208	8.530901	0.982418
Exogenous-only	Decision Tree	12.448	6.896	9.401441	8.495987	0.981417
Exogenous-only	Linear Regression	12.839	7.464	12.962123	10.413592	0.980230
Exogenous-only	MLP	12.64	8.88	14.635094	11.771445	0.980688
Exogenous-only	Tweedie GLM	42.363	28.958	61.071855	32.878515	0.784759

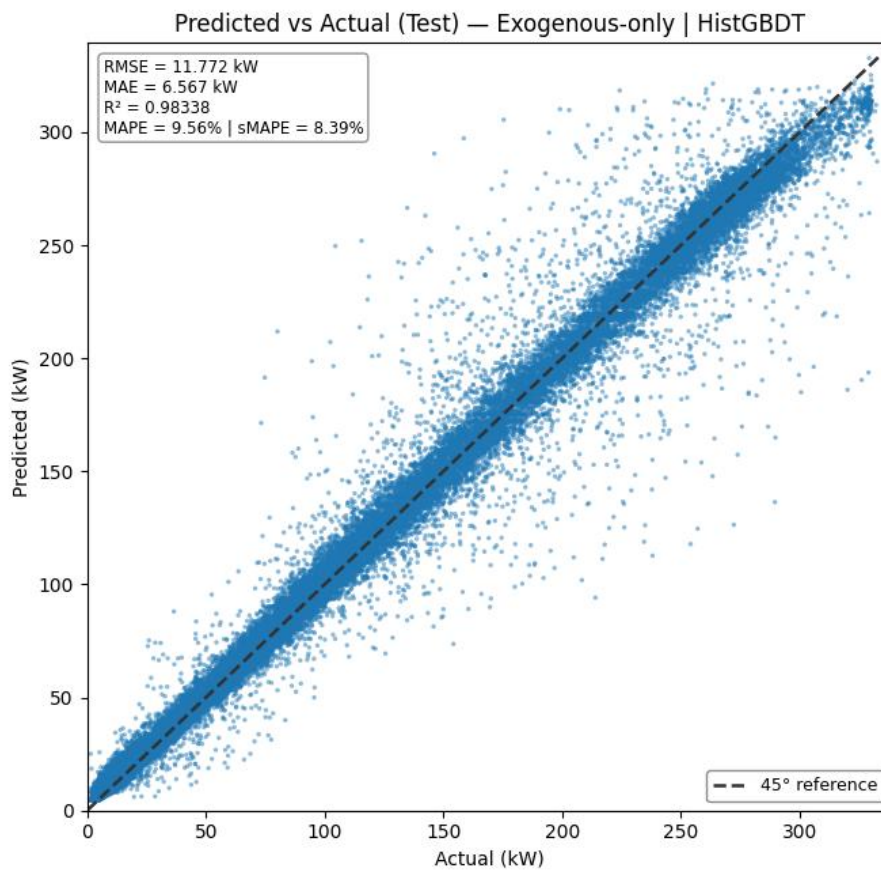


Figure 5. Predicted vs. Actual on the Test Set — Scenario A: HistGBDT

5.3. Scenario B: Operational Tracking Performance

When electrical telemetry (DC and three-phase AC currents) is incorporated alongside meteorological variables, model performance improves markedly across all evaluated methods. This improvement reflects the strong physical linkage between electrical currents and instantaneous AC power output.

As reported in Table 6, the HistGBDT model again delivers the best overall performance (RMSE = 1.84 kW, MAE = 1.12 kW, $R^2 = 0.9996$), closely followed by XGBoost. The MLP model also benefits markedly from the inclusion of electrical features, achieving RMSE = 2.31 kW and $R^2 = 0.9994$. However, this gain should be interpreted cautiously, as it reflects enhanced contemporaneous tracking rather than forward-looking forecasting (Porawagamage et al., 2024).

Table 6. Scenario B (Exogenous + Currents) test performance

Scenario	Model	RMSE	MAE	MAPE (%)	sMAPE (%)	R^2
Exogenous+Currents	HistGBDT	1.844902	1.121128	1.143953	1.137031	0.999592
Exogenous+Currents	XGBoost	1.899313	1.160020	1.144693	1.137617	0.999567
Exogenous+Currents	MLP	2.308639	1.445145	2.010189	2.025549	0.9993607
Exogenous+Currents	Linear Regression	2.486102	1.606102	2.340750	2.180932	0.999259
Exogenous+Currents	Decision Tree	2.563009	1.502813	1.389074	1.386648	0.999212
Exogenous+Currents	SVR (RBF)	2.736564	1.449176	2.623016	2.278583	0.999102
Exogenous+Currents	Tweedie GLM	34.214707	22.739552	49.448920	27.706796	0.859598

Compared with Scenario A, the inclusion of current measurements reduces RMSE and MAE by approximately 84% and 83%, respectively. These results define an empirical upper bound for SCADA-based PV power estimation under synchronous sensing conditions (Porawagamage et al., 2024). Consequently, Scenario B is interpreted as an operational tracking (nowcasting) task, whereas Scenario A represents a more generalizable forecasting setting applicable to systems lacking electrical telemetry.

The predicted versus actual scatter in Figure 6 shows near-perfect alignment along the 45° reference line, indicating negligible bias and outstanding fit under the tracking-oriented scenario.

5.4. Thermal Derating Analysis

Temperature-induced derating represents a critical nonlinearity in PV system behavior, directly affecting energy yield and operational reliability. To quantify thermal sensitivity, two complementary approaches were employed: (i) banded ordinary least squares (OLS) regressions under high-irradiance conditions and (ii) a semiparametric regression combining spline terms for irradiance with a linear temperature component.

As shown in Figure 7, OLS-based temperature slopes decline monotonically with increasing irradiance. At moderate irradiance levels (400–700 $\text{W}\cdot\text{m}^{-2}$), slopes are near zero or slightly positive, whereas beyond 800 $\text{W}\cdot\text{m}^{-2}$ they become distinctly negative (-0.1 to $-0.45 \text{ kW}\cdot^\circ\text{C}^{-1}$), indicating progressive thermal derating at elevated flux.

The semiparametric regression model provides a unified estimate of temperature sensitivity after accounting for nonlinear irradiance effects (Figure 8). The model explains over 98% of the variance ($R^2 = 0.982$) and yields a statistically significant temperature coefficient ($\beta_t = -0.192 \text{ kW} \cdot ^\circ\text{C}^{-1}$, 95% CI: -0.227 to -0.158).

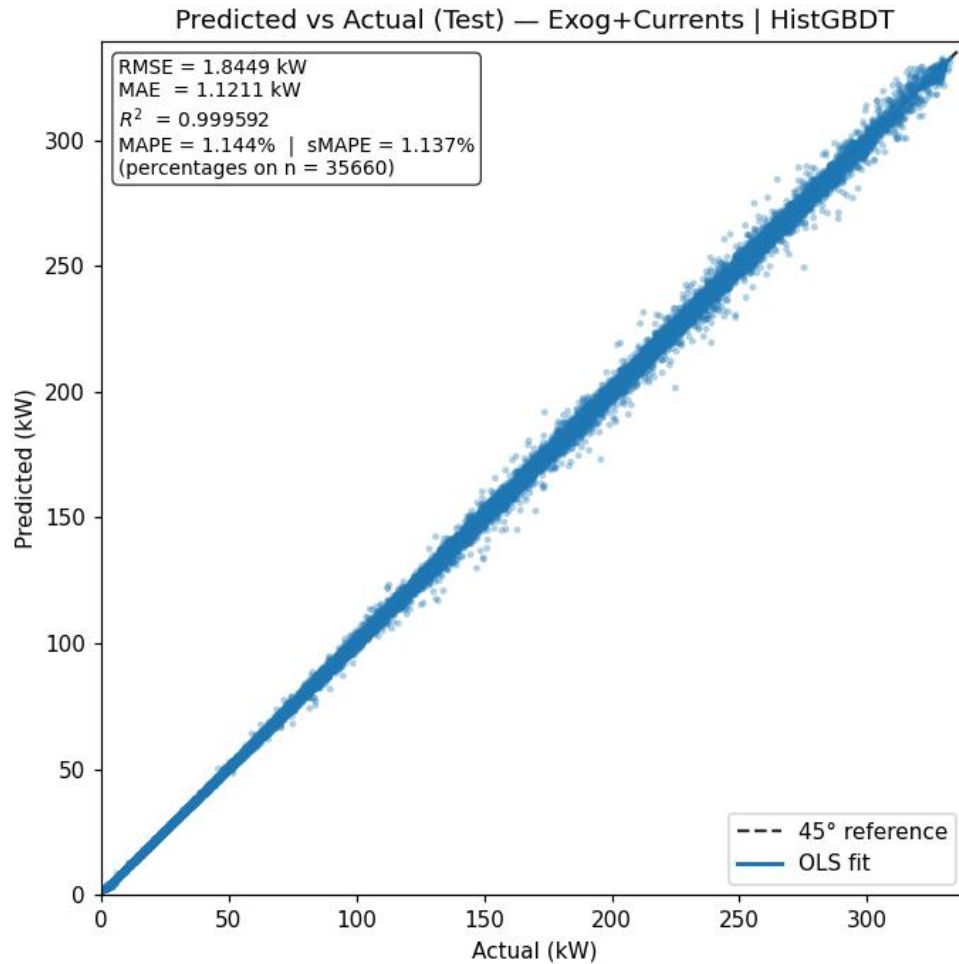


Figure 6. Predicted vs. Actual on the Test Set —Scenario B: HistGBDT

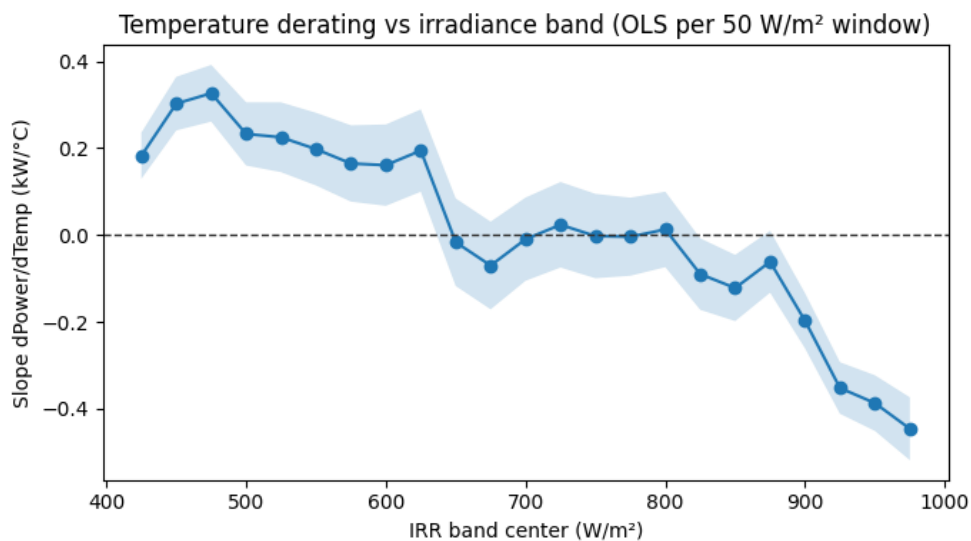


Figure 7. HistGBDT High-IRR OLS slopes by irradiance band

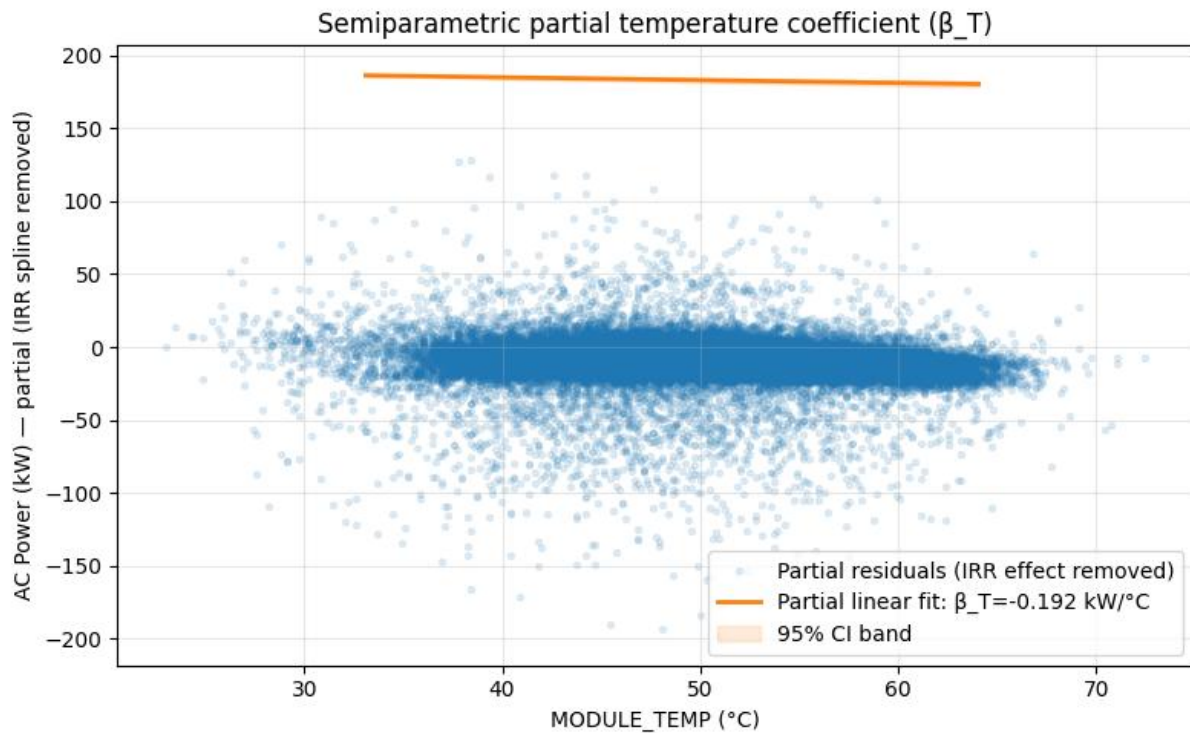


Figure 8. Semiparametric partial temperature coefficient (β_T)

This estimate represents the net temperature effect after accounting for irradiance and is fully consistent with established crystalline silicon PV physics. Table 7 summarizes both approaches, highlighting the improved stability and interpretability of the semiparametric formulation relative to per-band OLS fits.

Table 7. Summary of thermal-derating coefficients

Model	β_T (kW·°C ⁻¹)	95 % CI (lower–upper)	R ²
High-IRR OLS (banded, 600–1000 W·m ⁻²)	−0.07 → −0.45 (approx.)	band-specific (see Fig. 4A)	≤ 0.03
Semiparametric (spline IRR + linear Temp)	−0.192	(−0.227 to −0.158)	0.982

5.5. Explainability Analysis (SHAP, PDP, ICE)

Post-hoc explainability was performed using SHAP and complementary Partial Dependence (PDP) and Individual Conditional Expectation (ICE) plots. These analyses ensure both interpretability and physical plausibility of the learned mappings. For the exogenous-only configuration, Figures 9 and 10 show that irradiance (IRR) overwhelmingly dominates model output magnitude, confirming its central role in explaining AC power variability. Module temperature contributes a minor but consistent negative influence, while ambient temperature and wind speed exert negligible effects. When electrical features are introduced (Figures 11 and 12), phase-averaged AC current ($_AC_lavg$) and DC current emerge as dominant predictors, effectively absorbing irradiance effects through current–power coupling.

The PDP/ICE curves (Figures 13–14) visualize these transitions: under exogenous-only inputs, AC power increases near-linearly with IRR and declines slightly with module temperature, matching expected physical

behavior; when currents are included, PDP surfaces flatten, indicating that most variability is already explained by electrical telemetry.

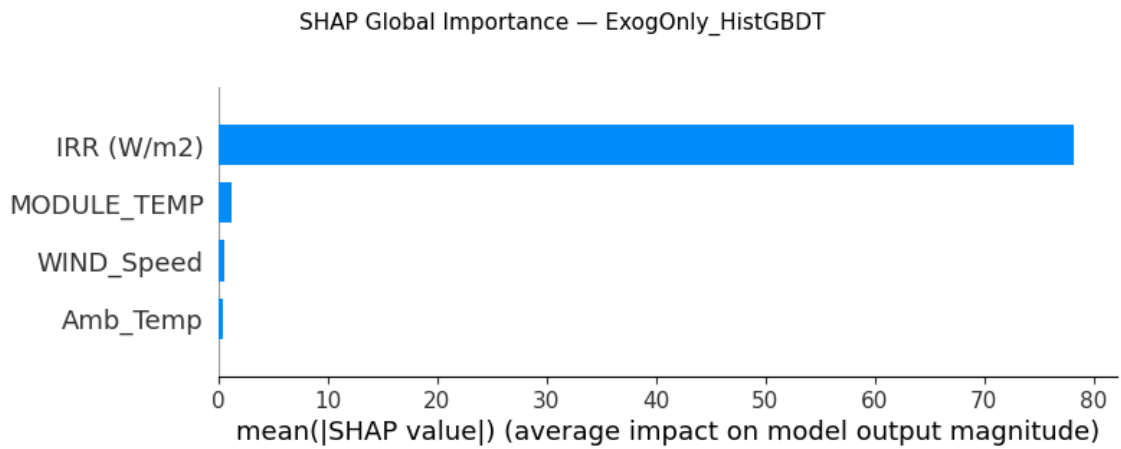


Figure 9. SHAP Global Importance (Exogenous-only | HistGBDT)

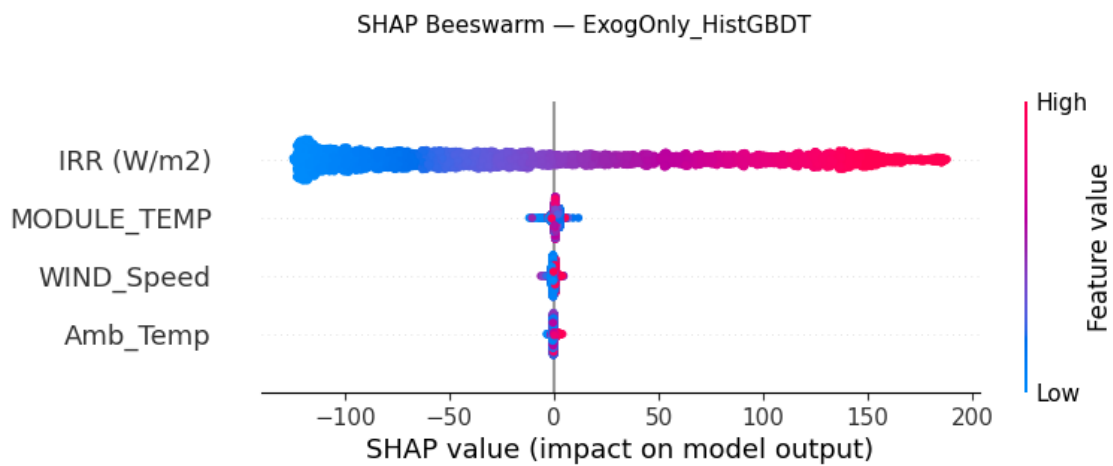


Figure 10. SHAP Beeswarm (Exogenous-only | HistGBDT)

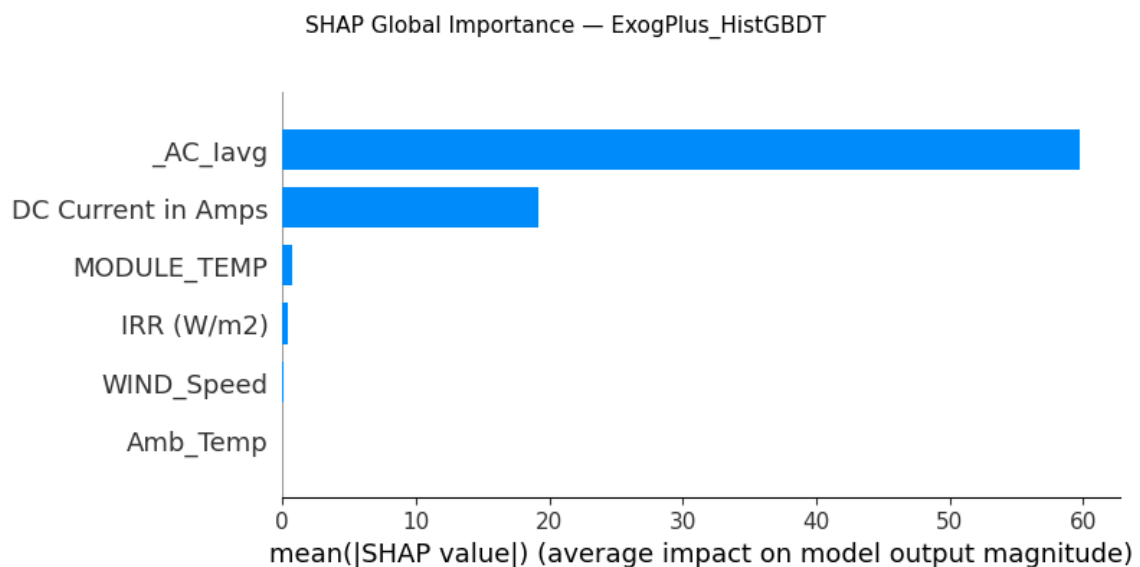


Figure 11. SHAP Global Importance (Exog + Currents | HistGBDT)

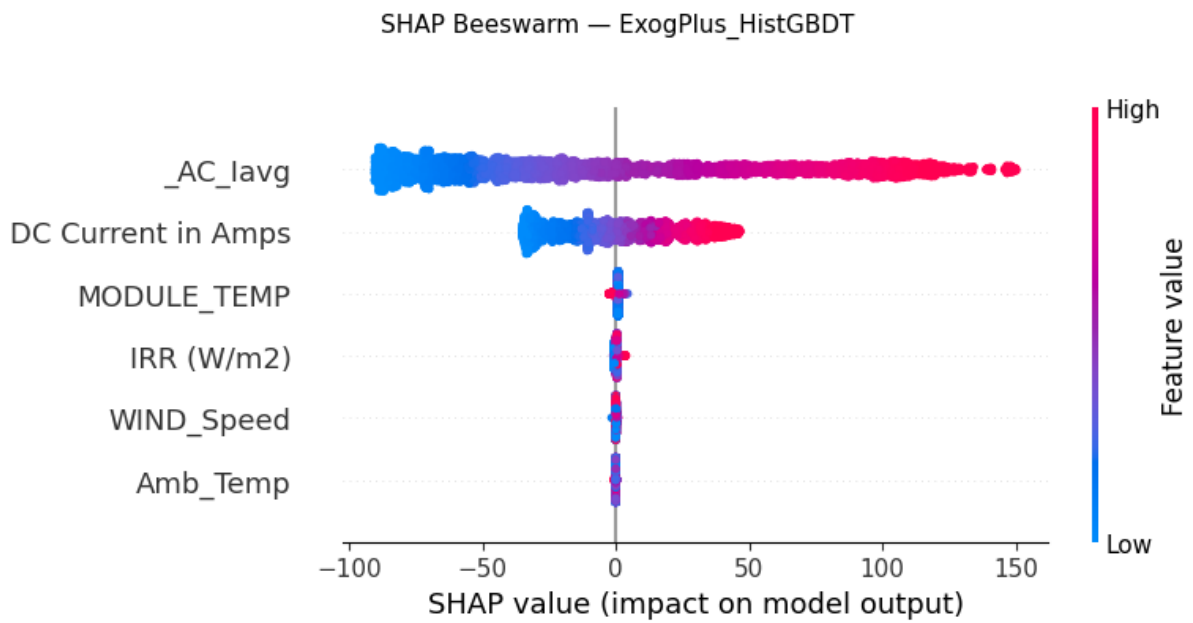


Figure 12. Beeswarm (Exog + Currents | HistGBDT)

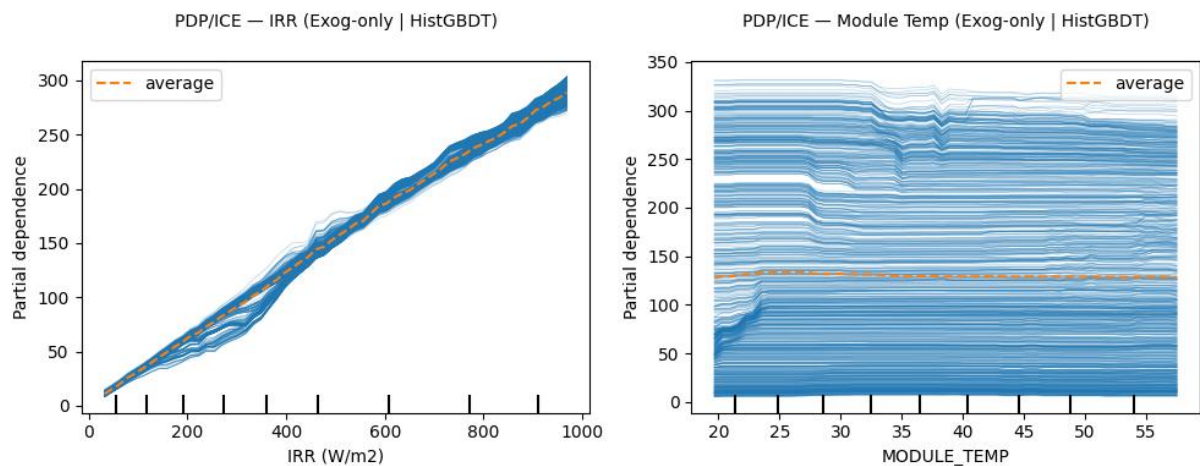


Figure 13. PDP/ICE (IRR and Module Temp, Exogenous-only | HistGBDT)

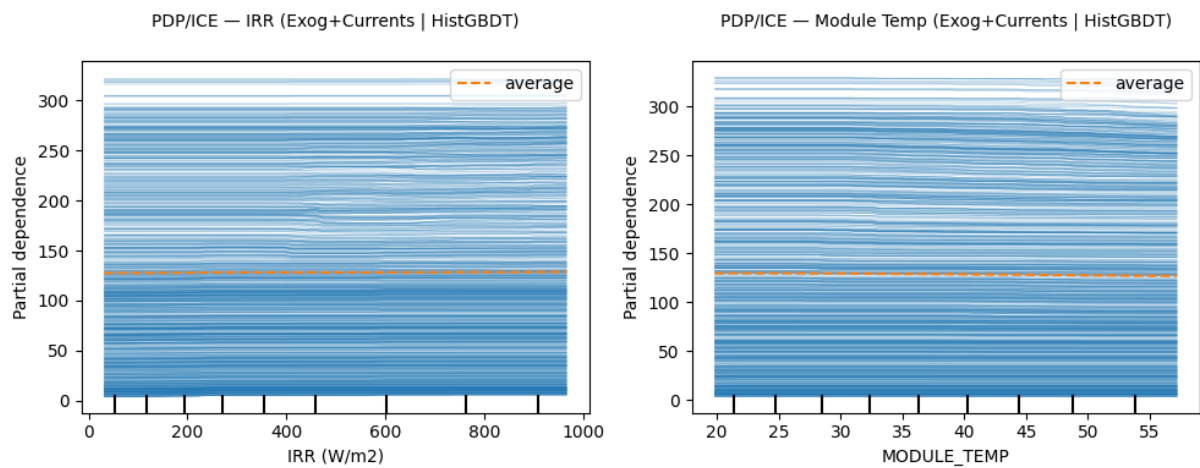


Figure 14. PDP/ICE (IRR and Module Temp, Exog + Currents | HistGBDT)

Collectively, these explainability diagnostics confirm that HistGBDT captures physically meaningful relationships irradiance as the main exogenous driver, temperature as a consistent derating factor and currents as operational proxies. This combination ensures both high accuracy and transparency, making the proposed pipeline suitable for real-world deployment and grid-integration decision support.

5.6. Comparative Analysis with Previous Studies

Comparing PV power forecasting performance across the literature is inherently challenging due to substantial differences in data sources, forecast horizons, input feature availability and evaluation protocols. In particular, reported accuracy metrics are often not directly comparable when studies differ in whether they address true exogenous forecasting or contemporaneous power tracking using synchronous electrical measurements.

To address this issue and respond to reviewer concerns, the comparative analysis in this study is deliberately restricted to methodologically comparable works. Specifically, only studies that (i) rely on operational or SCADA-like PV datasets, (ii) focus on short-term forecasting or short-horizon estimation tasks and (iii) report commonly used error metrics such as RMSE, MAE, or MAPE are considered. Studies employing fundamentally different horizons (e.g., day-ahead forecasting), aggregated regional data, or incompatible evaluation settings are intentionally excluded to avoid misleading comparisons.

Rather than ranking models based on absolute error values, the comparison emphasizes contextual alignment, including task definition, feature composition and operational intent. Prior studies frequently report strong predictive performance by incorporating electrical variables measured synchronously with the target power output. While such configurations improve numerical accuracy, they correspond to operational tracking (nowcasting) rather than forward-looking forecasting. In contrast, the present study explicitly distinguishes between these two settings, enabling results to be interpreted within their appropriate operational context.

Table 8 summarizes selected recent studies that satisfy the above comparability criteria. The table highlights how differences in data type, task formulation and feature availability influence the interpretation of reported performance metrics. Within this framework, the results obtained in this study are consistent with previously reported ranges under similar conditions. The primary contribution lies not in claiming superior accuracy, but in providing a leakage-aware and operationally realistic basis for comparison that clarifies how evaluation design shapes reported performance.

5.7. General Discussion and Practical Implications

This study set out to clarify how feature availability, evaluation design and operational intent shape reported performance in PV power modeling. The results demonstrate that predictive accuracy cannot be interpreted independently of the modeling scenario, particularly with respect to whether the task corresponds to true exogenous forecasting or contemporaneous operational tracking.

A key observation is the pronounced performance gap between Scenario A (exogenous-only inference) and Scenario B (current-augmented tracking). While the inclusion of electrical current measurements leads to a pronounced improvement in numerical accuracy, such gains primarily reflect access to target-adjacent information and therefore correspond to nowcasting rather than forward-looking prediction. This distinction is frequently blurred in the literature, contributing to optimistic performance claims that may not translate to real-world forecasting use cases (Porawagamage et al., 2024). By explicitly separating these scenarios, the present work provides a clearer and more operationally meaningful framework for interpreting reported accuracy.

Table 8. Comparison with selected PV power forecasting studies under comparable settings

Study	Data type	Horizon / Task	Input features	Metrics reported	Notes on comparability
(Di Leo et al., 2025)	Operational PV (review)	Short-term PV forecasting	Meteorological + electrical	RMSE, MAE, MAPE (reported across studies)	Highlights variability due to evaluation design
(Piantadosi et al., 2024)	Utility-scale PV	Short-term forecasting	Meteorological only	RMSE, MAE	Deep learning; exogenous-only
(Xie et al., 2025)	Large-scale PV plant	Short-horizon estimation	Meteorological + currents	RMSE	Tracking-oriented setup
(Fu et al., 2025)	Building-integrated PV	Short-term forecasting	Meteorological + shading	RMSE, MAE, SHAP	Strong interpretability focus
This study (Scenario A)	SCADA PV plant	Exogenous inference	Meteorological only	RMSE, MAE, R ²	Leakage-aware forecasting
This study (Scenario B)	SCADA PV plant	Operational tracking	Meteorological + currents	RMSE, MAE, MAPE	Nowcasting; not forward forecasting (MAPE for y > 0.5 kW)

From a physical perspective, the analysis demonstrates that leakage-aware, exogenous-only models can recover physically consistent relationships despite the absence of electrical measurements. Explainability analyses and the thermal derating assessment indicate that irradiance remains the dominant driver of PV output, while module temperature exerts a systematic negative influence under high-irradiance operating conditions. The semiparametric temperature coefficient estimated in this study provides a quantitatively interpretable measure of thermal sensitivity that is consistent with established crystalline silicon PV behavior and highlights the value of integrating physical reasoning into data-driven evaluation.

The practical implications of these findings depend strongly on the intended deployment context. For grid operation and short-term scheduling, where forecasts are required several minutes ahead and synchronous electrical measurements may not be available at prediction time, exogenous-only models such as those evaluated in Scenario A offer a more realistic representation of operational conditions. In contrast, Scenario B models are better suited for applications such as monitoring, fault detection and performance tracking, where the objective is accurate contemporaneous estimation rather than forecasting.

Forecast horizon realism is another important consideration. The present study focuses on instantaneous or near-instantaneous estimation using high-frequency SCADA data. Performance at longer horizons (e.g., 5–30

minutes ahead) is expected to degrade as temporal uncertainty increases and meteorological forecast errors accumulate. Although such horizons are beyond the scope of the current analysis, the leakage-aware framework adopted here is directly applicable to future work addressing explicit multi-step forecasting problems.

Several limitations should be acknowledged. The analysis is based on data from a single PV plant and generalization to other installations may depend on site-specific characteristics such as climate, system configuration and sensor quality. In addition, while a shallow neural network baseline was included, the study does not aim to exhaustively evaluate deep learning architectures. Instead, the emphasis is placed on methodological clarity, interpretability and operational realism. Future research could extend this framework to recurrent or transformer-based models under comparable leakage-aware evaluation settings.

Overall, this work underscores that improvements in reported accuracy must be interpreted in light of feature availability, evaluation design and deployment context. By emphasizing operational realism, physical consistency and transparent experimental protocols, the proposed framework provides practical guidance for both researchers and practitioners seeking to develop and deploy PV power models in real-world environments.

6. CONCLUSION

This study examined PV power modeling under a leakage-aware and operationally realistic framework, with particular emphasis on how feature availability, evaluation design and task definition influence reported performance. By explicitly distinguishing between exogenous forecasting and contemporaneous operational tracking, the work addresses a persistent source of ambiguity in the PV forecasting literature, where these two tasks are frequently conflated.

The results indicate that models relying exclusively on meteorological inputs can achieve physically consistent and interpretable performance when evaluated under realistic assumptions. Although the inclusion of electrical current measurements leads to noticeably lower numerical errors, such improvements reflect access to synchronous, target-adjacent information and therefore correspond to operational tracking rather than forward-looking forecasting. Recognizing this distinction is essential for the correct interpretation of model accuracy and for aligning modeling approaches with real-world deployment constraints.

The integration of post-hoc explainability analyses further demonstrates that leakage-aware, exogenous-only models recover relationships that are consistent with established PV physics. Irradiance emerges as the dominant driver of power output, while temperature-induced derating becomes evident at high operating levels. These findings support the use of interpretable, tree-based ensemble models as practical tools for performance estimation and diagnostic analysis, provided that feature selection and evaluation protocols are carefully designed.

From an operational standpoint, the proposed framework offers practical guidance for selecting appropriate modeling strategies based on application context. Exogenous-only models are more suitable for short-term forecasting and grid-oriented decision making, where predictions must be generated without access to synchronous electrical measurements. In contrast, current-augmented models are better suited for monitoring, fault detection and real-time performance tracking within operational PV plants.

Several limitations should be acknowledged. The analysis is based on data from a single PV installation and broader generalization may require validation across diverse climatic regions and system configurations. In addition, while a shallow neural network baseline was included, the study does not aim to exhaustively evaluate deep learning architectures. Future work may extend the proposed leakage-aware framework to explicit multi-step forecasting horizons and to recurrent or transformer-based models under comparable operational constraints.

In conclusion, this work emphasizes that improvements in predictive accuracy must be interpreted in light of feature availability, task definition and deployment context. By prioritizing evaluation realism, physical consistency and transparent experimental design, the study provides a methodologically grounded and practically relevant perspective for the development and assessment of PV power models in real-world applications.

6.1. Limitations and Future Work

Despite the strong performance of the proposed approach, several limitations and research opportunities remain.

First, the enhanced accuracy observed in Scenario B partially arises from target-adjacent current measurements that may not always be available in predictive settings.

Future studies should therefore evaluate generalization under purely exogenous inputs or when current signals are delayed or missing.

Second, this analysis focused on a single utility-scale PV plant.

Extending validation across multiple sites and diverse climatic conditions would help assess the model's spatial generalizability and robustness to regional variations.

Third, the current framework operates under an offline learning paradigm.

Integrating adaptive or online learning mechanisms could enhance resilience against nonstationary weather dynamics, sensor drift and seasonal variability.

Fourth, incorporating physics-informed neural networks (PINNs) or spatio-temporal ensemble hybrids in future work could further improve both explainability and cross-domain transferability.

Future research will thus focus on scaling the proposed explainable ensemble to heterogeneous PV fleets, integrating cloud-cover maps and satellite irradiance data and developing hybrid digital twins that combine data-driven learning with physical simulation.

Such extensions will enhance the operational readiness of explainable AI for renewable energy forecasting and smart-grid optimization.

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AUTHOR CONTRIBUTIONS

All human authors have read and legally accepted the final version under the CC BY-SA 4.0 framework.

CONFLICT OF INTEREST

The author declares no conflict of interest.

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